JONATHAN TURNER

Software Engineer

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Summary

Software engineer with a Computer Science degree concentrated in Artificial Intelligence and hands-on experience building real-time computer vision, autonomy, and automation systems. Comfortable working across the stack, from Dockerized Python microservices and edge AI inference to embedded robotics validation, with a track record of triaging complex failures, root-causing defects, and shipping fixes alongside software teams.

Education

Kennesaw State University Kennesaw, GA

Bachelor of Science in Computer Science with a Concentration in Artificial Intelligence Aug 2021–May 2025

GPA: 3.88 | *Magna Cum Laude*

Relevant Coursework: Algorithms & Data Structures, Machine Learning, Operating Systems, Computer Organization & Architecture, Artificial Intelligence

Professional Experience

Serve Robotics Atlanta, GA

Lead Robotics Technician June 2025–Present

- Debug software defects at the system and application level across autonomous robot subsystems, contributing targeted fixes and validating patches across hardware revisions in collaboration with the software engineering team.
- Author and maintain Python and Bash automation for test execution, telemetry parsing, log analysis, and fault reproduction, reducing manual debugging effort and scaling test coverage across the fleet.
- Triage 5–10 hardware and software defects daily, performing root cause analysis and producing structured incident reports; coordinate next-steps and resolution with software engineering. Identified a firmware initialization defect causing fleet-wide boot failures, averting significant downtime.
- Execute software validation and QA across robot subsystems, verifying firmware and application-layer behavior against functional specifications and release acceptance criteria.
- Perform board bringup and embedded hardware validation on autonomous delivery robot platforms, diagnosing power rail, signal integrity, and firmware initialization failures.
- Sustain greater than 95% fleet health and operational uptime through preventive maintenance and field servicing of autonomous robot fleets.

Wyzant Tutoring Canton, GA

Undergraduate Computer Science Tutor Apr 2024–June 2025

- Delivered one-on-one instruction in Algorithms & Data Structures, Machine Learning, Operating Systems, and Computer Organization & Architecture, adapting strategies to diverse learning styles.
- Built customized practice exercises and review materials to reinforce core concepts; maintained a consistent 5-star rating across the platform.

Best Buy Geek Squad Kennesaw, GA

Consultation & Repair Technician Apr 2021–June 2025

- Diagnosed and repaired complex hardware and software issues across desktops, laptops, and peripherals using specialized diagnostic tooling and structured troubleshooting.
- Documented service workflows, resolution steps, and best practices in an internal knowledge base, improving consistency and communication across cross-functional teams.

Project Experience

IcarusEye: Edge AI Vision Pipeline Aug 2025–Present

- Architected a modular, production-grade real-time AI vision pipeline for NVIDIA Jetson and other edge devices: ingests HDMI drone/FPV video via V4L2 capture, runs YOLO-based inference via TensorRT (INT8), and streams annotated output over RTP for low-latency detection.
- Designed a microservices architecture with Redis pub/sub messaging and Docker containerization using architecture-optimized base images for hardware-accelerated inference; built a factory-pattern capture abstraction supporting V4L2, RTSP, and file sources via config-only switching.
- Engineered a field-deployable power system (lithium pack with integrated BMS) and custom power/data interconnect, surfacing telemetry into the pipeline dashboard for real-time cell voltage, current draw, and state-of-charge monitoring.

Self-Hosted Infrastructure Platform 2024–Present

- Architected and operate a multi-node, self-hosted infrastructure platform spanning Docker service orchestration, Apache/Caddy reverse proxies, mesh VPN networking, and centralized storage, all managed as code with dependency-aware startup and automated updates.
- Deployed a local AI stack (Ollama, Open WebUI, self-hosted search) on Jetson hardware and a distributed transcoding cluster, applying the same compute, storage, and networking patterns used in production cloud environments.

GSO Website Unification Jan 2025–May 2025

- Led consolidation of multiple Georgia Symphony Orchestra domains into a single WordPress platform with custom themes, plugins, and API integrations.
- Delivered responsive, front-end design and a streamlined ticketing UX for consistent branding and ease of use.

Technical Skills

- **Languages:** Python, Java, Bash/Shell Scripting, C#, MATLAB
- **CS Fundamentals:** Algorithms & Data Structures, Object-Oriented Programming, Operating Systems, Computer Organization & Architecture
- **AI/ML & Vision:** Computer Vision, YOLO, TensorRT (INT8), Edge AI Inference, Telemetry & Sensor Data Processing
- **Cloud & Infrastructure:** Docker & Docker Compose, Microservices, Redis, Linux, Reverse Proxies, Mesh VPN/Networking, Git/GitHub
- **Testing & Validation:** Software QA, Fault Analysis, Root Cause Analysis, Test Automation, Telemetry/Log Analysis, Fault Reproduction
- **Hardware & Embedded:** Board Bringup, Embedded Diagnostics, Firmware Validation, Power Rail & Signal Integrity Analysis
- **Front-End:** Vue/Vite, WordPress, Responsive Design

Socials

LinkedIn — LTurnerJonathan **GitHub** — JonathanLTurner03 **Website** — atlantissrv.com